Draft version September 25, 2026
Typeset using LaTeX **default** style in AASTeX7.0.2

Two Transit-like Signals in TESS Photometry of the Nearby K Dwarf TIC 4206066

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ABSTRACT

I report two transit-like signals in TESS photometry of the late-K dwarf TIC 4206066 (StKM 1-561), found in Sector 98 and traced back to Sectors 6 and 32. The 3.18 d signal (23 transits, ≈500 ppm) corresponds to a $1.4 \pm 0.1\,R_\oplus$ planet if it orbits the target. Vetting, localization, and Gaia astrometry, photometry, and radial velocities leave a planet transiting an unresolved bound companion as the main alternative host scenario; TRICERATOPS gives FPP = 0.03–0.04, mostly from companions these data disfavor. The lower-significance 11.13 d signal (≈$2.2\,R_\oplus$ if on the target; ≈ 3σ in a post hoc test) is reported with a tentative ephemeris.

1. TARGET AND DATA

TIC 4206066 (Gaia DR3 3220388198192519424; $T = 9.68$) lies at 35.6 pc (Gaia Collaboration et al. 2016; Gaia Collaboration et al. 2023). From $M_{K_s} = 4.94$, the relations of A. W. Mann et al. (2015, 2019) give $R_\star = 0.623 \pm 0.019\,R_\odot$, $M_\star = 0.606 \pm 0.015\,M_\odot$, and $\rho_\star = 2.51^{+0.25}_{-0.22}\,\rho_\odot$. I searched Sector 98 Quick-Look Pipeline (QLP) light curves (C. X. Huang et al. 2020) of 126,246 stars ($T \leq 11.5$) with box least squares (G. Kovács et al. 2002). Both passed all LEO-Vetter flux-level tests (M. Kunimoto et al. 2025) but their multiple-event statistics (10.6, 11.0) fell below the shortlist threshold (15), so I examined them post hoc. I use TESS (G. R. Ricker et al. 2015) light curves from TESS-SPOC (Sector 6, 30 min PDCSAP; D. A. Caldwell et al. 2020), SPOC (Sector 32, 2 min PDCSAP; J. M. Jenkins et al. 2016), and QLP (Sector 98, 200 s, TGLC-based; T. Han & T. D. Brandt 2023; G. Petitpas et al. 2026), removing all flagged Sector 98 cadences.

2. THE 3.18 DAY SIGNAL

An epoch search at the Sector 98-only period recovers the signal in Sectors 6 and 32 (false-alarm probability $\lesssim 1\%$ per sector, $\lesssim 0.1\%$ jointly, trials included). Transit times from a fixed-shape batman template (L. Kreidberg 2015) with injection-calibrated errors give $P = 3.182785 \pm 0.000006$ d and $T_0 = 2460998.736 \pm 0.003$ $\mathrm{BJD_{TDB}}$ from the 19 transits with ≥70% coverage, S/N ≥ 2, and ≥15 min of post-egress data ($\chi^2/\mathrm{dof} \approx 9.5/17$; no evidence of timing variations). An exposure-integrated emcee fit (D. Foreman-Mackey et al. 2013) of all 23 transits (circular orbit; A. Claret 2017 limb darkening; Mann density prior) gives $b = 0.35^{+0.10}_{-0.14}$, $T_{14} = 1.87 \pm 0.05$ h, and $R_p/R_\star = 0.0204 \pm 0.0010$, so $R_p = 1.4 \pm 0.1\,R_\oplus$. With free density, $\rho_\star \lesssim 3.6\,\rho_\odot$ (97.7%). Per-sector and odd/even depths agree, the phase-0.5 depth is -20 ± 32 ppm, and a V shape is disfavored ($\Delta\chi^2 = 32$; ≥ 21 with red noise). Without Sector 6 these values change by $\lesssim 1\sigma$.

3. THE 11.13 DAY SIGNAL

A lower-significance 11.13 d signal (≈1000 ppm; $R_p \approx 2.2\,R_\oplus$ if on the target; offset $(+1 \pm 3, -1 \pm 3)''$) has two clean Sector 98 transits, which alone do not establish it, and dips consistent with its period in Sector 6 (marginally in Sector 32), at ≈ 3σ (false-alarm probability ≈ 2×10^{-3}) in a recovery test defined after inspecting the archival data, where an unanchored search had failed. Its Sector 32 events abut 3.18 d transits; assuming the archival dips are real, its ephemeris is $P = 11.13274 \pm 0.00003$ d and $T_0 = 2461000.865 \pm 0.003$ $\mathrm{BJD_{TDB}}$.

4. LOCALIZATION AND FALSE POSITIVES

SPOC (Sector 32, pointing-decorrelated) and TESScut (Sector 98; C. E. Brasseur et al. 2019) difference images fitted with the TESS pixel response function (K. J. Bell & M. E. Higgins 2022) place the 3.18 d source at $(+5 \pm 3, 0 \pm 3)''$

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from the target ($\approx 2\sigma$; $\leq 2.6\sigma$ across Sector 98 reductions). Sector 6 difference images are unusable (transits lie 0.4–3.5 h from momentum dumps). All Gaia DR3 stars able to produce the depth lie $\gtrsim 43''$ ($> 10\sigma$) away. Gaia DR3 gives RUWE = 1.21 and radial-velocity scatter ($p = 4 \times 10^{-5}$) that are elevated (origin unknown) but below the RUWE (1.4) and gof (3.0 vs. 4) thresholds, no non-single-star solution or other source within $10''$, and radial velocities ($K < 1.2\,\mathrm{km\,s^{-1}}$) that exclude an eclipsing stellar companion of the target. Against the 0.05–0.07 mag scatter of comparable stars, the XP overluminosity of Z. Way et al. (2026), +0.003 mag, disfavors Gaia-blended ($\lesssim 0.2''$) companions above $\approx 0.38\,M_\odot$ (2σ; M. J. Pecaut & E. E. Mamajek 2013); at 0.2–2″ such companions would be expected to raise the BP/RP flux excess (M. Riello et al. 2021), which is normal. For the 3.18 d duration, a circular orbit needs a companion host above $\approx 0.52\,M_\odot$; under the adopted companion model, 0.15–0.38 $M_\odot$ hosts would require $e \gtrsim 0.2$–0.6.

TRICERATOPS 1.1.0 (S. Giacalone et al. 2021), with Sector 32+98 folds (5–10 min bins) and unresolved companions at 0.1–2.2″, gives FPP = 0.03–0.04 (4–20 runs of 10^6 draws), nearly all from bound companions, about 90% from those above $0.38\,M_\odot$; excluding these (post hoc) gives FPP $\approx$ 0.002–0.007. The default FPP exceeds the TRICERATOPS validation threshold (FPP < 0.015, NFPP $< 10^{-3}$; here NFPP $< 10^{-6}$), so I claim no validation.

5. STATUS

The 3.18 d signal is a planet candidate whose main remaining false-positive channel needs high-resolution imaging and radial velocities; the 11.13 d signal is a tentative ($\approx 3\sigma$, post hoc) signal. Sector 110 (2026 October 31–November 26) spans eight or nine 3.18 d transit times (one in a scheduled data gap) and three 11.13 d ones. As of 2026 September 25 (UTC), I found no prior report of either signal in ExoFOP, TCE tables, the NASA Exoplanet Archive, candidate catalogs, SIMBAD, VizieR, ADS, or arXiv.

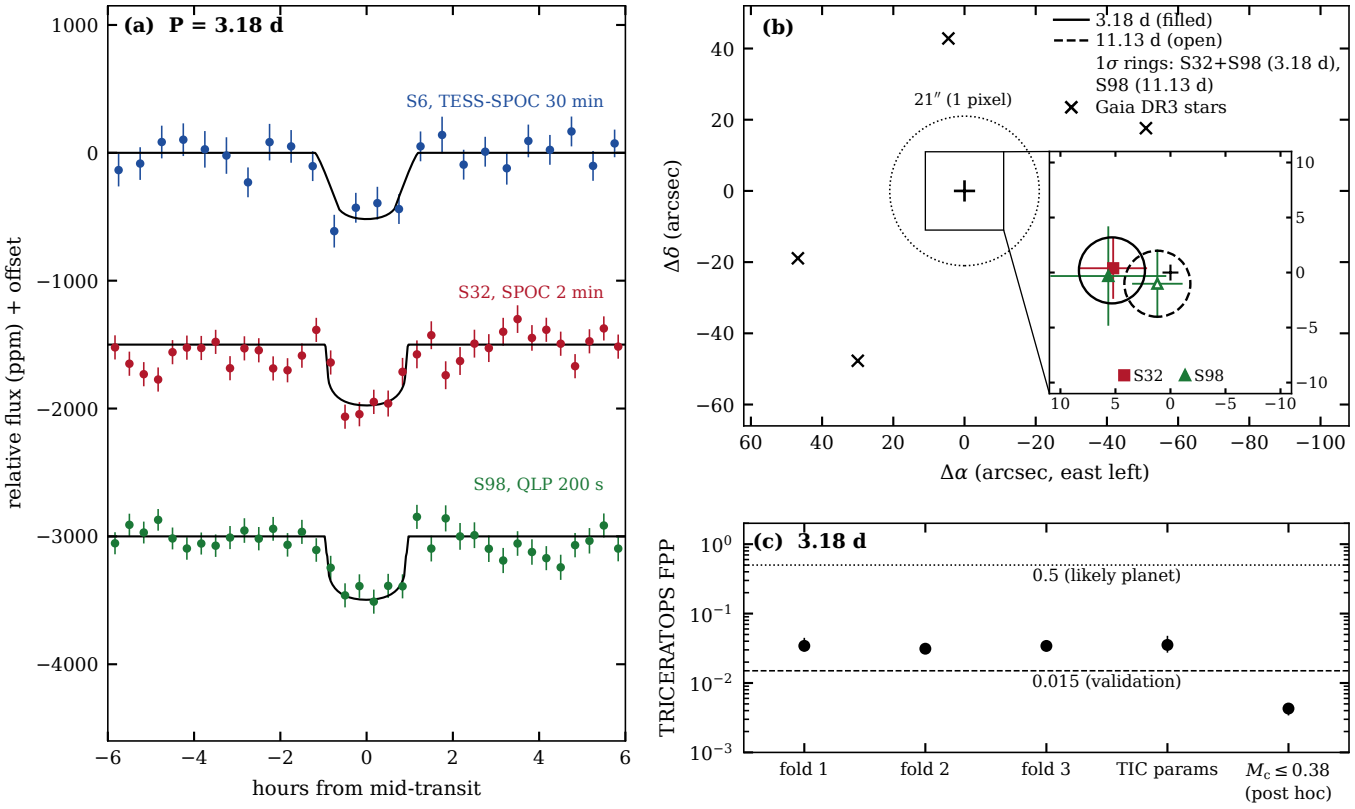


Figure 1. (a) 3.18 d transits per sector (binned; 11.13 d and one Sector 98 edge transit removed) with templates. (b) Difference-image offsets (inset: central 22″) with 1σ rings (3.18 d: Sectors 32+98; 11.13 d: Sector 98), the target (+), and Gaia DR3 stars able to produce the 3.18 d depth (crosses; a fifth at 84″). (c) TRICERATOPS FPP (mean, SD, range) for three 6-min-bin folds, TIC parameters, and companions capped at $0.38\,M_\odot$ (post hoc; 5-min fold). Plotted with Python code by Claude Code.

ACKNOWLEDGMENTS

This paper includes data collected by the TESS mission. It uses data from the ESA mission Gaia, processed by the Gaia DPAC, which is funded by national institutions, in particular those participating in the Gaia Multilateral Agreement. Code, data and derivations behind the numbers in this Note are deposited at doi:10.5281/zenodo.22967099.

AI disclosure: I designed and directed the study, checked all results and figures, and drew the conclusions. Claude Code (Anthropic) assisted with code, computations, figures, references, and drafting under my direction; ChatGPT (OpenAI; GPT-5.6 Sol, GPT-6 Pro) was used for review. I take full responsibility for the work.

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